

Managing AWS Cloud Infrastructure Efficiently with Terraform

Client Overview: MeraDoc, Smile Foundation NGO & CloudStok

MeraDoc is a holistic, tech-enabled healthcare delivery platform in India, focused on elderly care services. **Smile Foundation NGO** works towards empowering underprivileged children, youth, and women in India through various educational and healthcare initiatives. Both organizations required scalable, secure, and efficient cloud infrastructure to support their growing operations.

Challenges

- **Manual infrastructure management** led to errors, delays, and difficulty scaling resources.
- Lack of **consistency** across environments (dev, staging, production).
- High costs due to **inefficient resource provisioning** and poor visibility into cloud usage.
- Slow and error-prone **deployment processes**.

Our Solution

OnEggy Technologies implemented a **fully automated infrastructure provisioning process** using **Terraform** to ensure:

- **Infrastructure as Code (IaC)**: All infrastructure is described and provisioned using Terraform scripts for version control and reproducibility.
- **Automated provisioning**: Automated the entire cloud environment setup, reducing the need for manual interventions and human errors.
- **Multi-environment consistency**: Ensured **seamless replication of environments** (dev, staging, prod) by managing configurations centrally using Terraform.
- **Cost-effective resource provisioning**: Optimized resource allocation and used **Terraform workspaces** to apply different configurations to various environments for better cost control.
- **Integrated security measures**: Included security best practices such as IAM roles, VPC configurations, and encryption standards directly into Terraform modules.

Results & Impact

- **Faster provisioning**: Reduced infrastructure setup time by **50%**.

- **Cost savings:** Streamlined resource allocation led to a **25% reduction in AWS costs**.
- **Improved security:** All infrastructure components are provisioned with secure defaults, reducing the risk of misconfigurations.
- **Consistency & scalability:** Replicated environments with ease, improving **scalability** for future growth and making **rollback** or updates effortless.
- **Faster deployments:** Terraform's **state management** and **modular approach** accelerated deployment and environment management.

Tech Stack

- Terraform | AWS EC2, S3, VPC | IAM | Security Groups | Kubernetes | ECS EKS

Conclusion

By leveraging **Terraform** for AWS cloud management, OnEggy Technologies provided **MeraDoc** and **Smile Foundation NGO** with more **scalable, reliable, and cost-effective infrastructure** that grows with their needs. Automation and consistency became key to empowering their teams and optimizing resources effectively.

Ready to automate your AWS infrastructure with Terraform?

Let's discuss how we can help you improve efficiency and scalability!